

Practice Task 4: Data Cleaning and Prep using Power Query

Objective:

To clean, transform, and prepare the dataset for analysis by creating a structured Data Model that can be visualized as an ER diagram.

Step 1: Inspect and Understand the Data

- Review each table for:
 - Missing values or blanks
 - Duplicate records
 - Incorrect data types (e.g., numbers stored as text)
- Rename columns clearly to make them consistent and meaningful.
 - ♦ This ensures that the dataset is clean and ready for further processing.

Step 2: Clean and Transform the Data

- Remove unnecessary columns that won't be used in analysis.
- Replace missing values where applicable (e.g., "N/A" → 0 or default value).
- Change column types as needed:
 - Text → Text
 - Dates → Date
 - Numbers → Whole Number or Decimal
- Split or merge columns if necessary (e.g., separate "Full Name" → "First Name" + "Last Name").

💡 Power Query allows transformations to be applied dynamically whenever the data is refreshed.

Step 3: Establish Relationships

- Define relationships between tables to organize the data for analysis.
- Drag foreign key columns to match primary key columns where applicable.
 - ♦ This creates a structured Data Model suitable for reporting or visualization.

Step 4: Visualize the Data Model

- Use **Diagram View** in Power Pivot to view the connections between tables.
- Arrange tables to clearly show how they relate.
 - ♦ This step confirms that the dataset is organized, consistent, and ready for analysis.

Step 5: Finalize and Use the Dataset

- The cleaned and structured dataset can now be used for:
 - PivotTables for analysis
 - Power BI dashboards for interactive visualization
 - Exporting to online ERD tools for reporting

	Order ID	Order Date	Arrival Date	Name	Department	Region	Payment Type	Revenue	Profit
2	25691	45359	45367	Bill SmITH	Cloud Tech	Texas	Credit Card	42845	8846
3	27870	45298	45382	KEN Singh	Strategy	New York	Card	34152	13489
4	46726	45482	45512	Harley Fritz	Strategy	New York	Credit Card	15829	67998
5	46726	45482	45512	Harley Fritz	Strategy	New York	Credit Card	15829	67998
3	35895	45504	45551	Nyla Novak	Operations	Florida	Card	50373	21335
7	27303	45290	45369	David Rasmussen	Operations	Florida	Transfer	46467	23668
3	25985	45566	45590	IVAN HINEY	Cloud Tech	Texas	Transfer	35036	9788
9	46060	45508	45522	JONha Ma	Cloud Tech	Texas	Card	35205	24643.5
0	32107	45325	45355	Jordan Boone	Cloud Tech	Texas	Bank Transfer	42246	29572.2
1	14750	45390	45442	Kylee Townsend	Cloud Tech	Texas	Bank Transfer	36906	20225
2	27691	45551	45609	Nora Rollins	Cloud Tech	Texas	Bank Transfer	14894	7988
3	47557	45386	45420	BRENDAN Wallace	Cloud Tech	Texas	Bank Transfer	38608	7721.6
4	46669	45312	45387	Conor Wise	Operations	Florida	Bank Transfer	46329.6	9265.92
5	38741	45558	45636	Steven MIChael	Big Data	California	Transfer	21889	13107.5
6	18517	45366	45413	Lucia Mckay	Big Data	California	Credit Card	18249	11589
7	15314	45627	45676	JoSE Roach	Big Data	California	Card	39047	20119
8	16569	45294	45321	Franklin WRIGT	Big Data	California	Transfer	26215	13255
9	32159	45495	45547	Alia Thornton	Operations	Florida	Transfer	43359	22488
0	12997	45481	45547	Denzel Flores	Operations	Florida	Card	39366	26798
1	33650	45476	45537	Conor Wise	Operations	Florida	Bank Transfer	40125	26800
2	28321	45469	45477	Steven MIChael	Big Data	California	Bank Transfer	48364	24659
3	35767	45359	45443	Bruno CordOVA	Big Data	California	Credit Card	15097	7899
4	38601	45214	45228	jaylynn napp	Big Data	California	Card	41034	29898
5	30084	45335	45384	Bruce RICH	Big Data	California	Transfer	27775	14697
6	24370	45658	45680	Arturo Moore	Big Data	California	Transfer	21230	13500
7	31377	45320	45354	Bryce Carpenter	Strategy	New York	Card	26577	13980
8	48068	45688	45697	Jaidyn ANDERSEN	Strategy	New York	Bank Transfer	17269	13815.2
9	39414	45416	45493	MARK WALM	Strategy	New York	Credit Card	27561	13233
0	21771	45596	45648	HARRY LEE	Strategy	New York	Card	41852	19880
1	27732	45506	45562	JOSH JOHNSON	Strategy	New York	Transfer	10615	4588

	A	B	C	D	E	F	G	H	I	J	K
1	Order ID	Order Date	Arrival Date	Numbers of Days Delivered	First Name	Last Name	Department	Region	Revenue	Profit	Profit Margin
2	25691	08/03/2024	16/03/2024		8 Bill	Smith	Cloud Tech	Texas	42845	8846	0.206465165
3	27870	07/01/2024	31/03/2024		84 Ken	Singh	Strategy	New York	34152	13489	0.394969548
4	46726	09/07/2024	08/08/2024		30 Harley	Fritz	Strategy	New York	15829	67998	4.295786215
5	35895	31/07/2024	16/09/2024		47 Nyla	Novak	Operations	Florida	50373	21335	0.423540389
6	27303	30/12/2023	18/03/2024		79 David	Rasmussen	Operations	Florida	46467	23668	0.509350722
7	25985	01/10/2024	25/10/2024		24 Ivan	Hiney	Cloud Tech	Texas	35036	9788	0.279369791
8	46060	04/08/2024	18/08/2024		14 Jonha	Ma	Cloud Tech	Texas	35205	24643.5	0.7
9	32107	03/02/2024	04/03/2024		30 Jordan	Boone	Cloud Tech	Texas	42246	29572.2	0.7
10	14750	08/04/2024	30/05/2024		52 Kylee	Townsend	Cloud Tech	Texas	36906	20225	0.548013873
11	27691	16/09/2024	13/11/2024		58 Nora	Rollins	Cloud Tech	Texas	14894	7988	0.536323352
12	47557	04/04/2024	08/05/2024		34 Brendan	Wallace	Cloud Tech	Texas	38608	7721.6	0.2
13	46669	21/01/2024	05/04/2024		75 Conor	Wise	Operations	Florida	46329.6	9265.92	0.2
14	38741	23/09/2024	10/12/2024		78 Steven	Michael	Big Data	California	21889	13107.5	0.598816757
15	18517	15/03/2024	01/05/2024		47 Lucia	Mckay	Big Data	California	18249	11589	0.635048496
16	15314	01/12/2024	19/01/2025		49 Jose	Roach	Big Data	California	39047	20119	0.515250852
17	16569	03/01/2024	30/01/2024		27 Franklin	Wright	Big Data	California	26215	13255	0.50562655
18	32159	22/07/2024	12/09/2024		52 Alia	Thornton	Operations	Florida	43359	22488	0.518646648
19	12997	08/07/2024	12/09/2024		66 Denzel	Flores	Operations	Florida	39366	26798	0.680739725
20	33650	03/07/2024	02/09/2024		61 Conor	Wise	Operations	Florida	40125	26800	0.667912773
21	28321	26/06/2024	04/07/2024		8 Steven	Michael	Big Data	California	48364	24659	0.509862708
22	35767	08/03/2024	31/05/2024		84 Bruno	Cordova	Big Data	California	15097	7899	0.523216533
23	38601	15/10/2023	29/10/2023		14 Jaylynn	Napp	Big Data	California	41034	29898	0.728615295
24	30084	13/02/2024	02/04/2024		49 Bruce	Rich	Big Data	California	27775	14697	0.529144914
25	24370	01/01/2025	23/01/2025		22 Arturo	Moore	Big Data	California	21230	13500	0.635892605
26	31377	29/01/2024	03/03/2024		34 Bryce	Carpenter	Strategy	New York	26577	13980	0.526018738
27	48068	31/01/2025	09/02/2025		9 Jaidyn	Andersen	Strategy	New York	17269	13815.2	0.8
28	39414	04/05/2024	20/07/2024		77 Mark	Walm	Strategy	New York	27561	13233	0.480134973
29	21771	31/10/2024	22/12/2024		52 Harry	Lee	Strategy	New York	41852	19880	0.475007168
30	27732	02/08/2024	27/09/2024		56 Josh	Johnson	Strategy	New York	10615	4588	0.432218559
31	38800	20/11/2024	06/12/2024		16 Mik	Naam	Strategy	New York	50763	26990	0.531686465

Practice
Order ID INTEGER
Order Date TIMESTAMP
Arrival Date TIMESTAMP
Numbers of Days Delivered INTEGER
First Name TEXT
Last Name TEXT
Department TEXT
Region TEXT
Revenue FLOAT
Profit FLOAT
Profit Margin FLOAT

Raw Data
Order ID INTEGER
Order Date INTEGER
Arrival Date INTEGER
Name TEXT
Department TEXT
Region TEXT
Payment Type TEXT
Revenue FLOAT
Profit FLOAT